

Some definitions

set is finite sets implemented as Lists.

$Subst := set (Var * exp)$

$Equation := set ((exp * exp) + (sexp * sexp))$

$Problem := set Equation$

$Tuple := Subst * Problem$

$-\langle _ \rangle : exp \rightarrow Subst \rightarrow exp$

$-\langle _ \rangle : sexp \rightarrow Subst \rightarrow sexp$

$-\langle _ \rangle : Problem \rightarrow Subst \rightarrow Problem$

$-\ @ _ : Subst \rightarrow Var \rightarrow exp$

$dom : Subst \rightarrow set Var$

$smatch : Tuple \rightarrow Tuple \rightarrow Prop$

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Congruence steps

$$\begin{aligned} \text{smatch_app_congr} : (s \approx t \approx s' \approx t') \in P \rightarrow \\ \text{smatch}(S, P) \text{ } (S, P + (s \approx s') + (t \approx t') \setminus (s \approx t \approx s' \approx t')) \end{aligned}$$

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σ -min steps

$$\begin{aligned} \text{smatch_subst_comp} : (s[\sigma \circ \tau] \approx s'[\rho]) \in P \rightarrow \\ \text{smatch } (S, P) \ (S, P + (s[\sigma][\tau] \approx s'[\rho]) \setminus (s[\sigma \circ \tau] \approx s'[\rho])) \end{aligned}$$

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Instantiation case

$$\begin{aligned} \textit{smatch_inst} : (s =_? \ X) \in P \rightarrow \\ X \notin \textit{lhvars_of} \ P \rightarrow \\ S' =_{\sigma_{\min}} S \gg \{(X, s)\} \rightarrow \\ \textit{smatch} \ (S, P) \ (S', (P \setminus (s =_? \ X)) \langle \{(X, s)\} \rangle) \end{aligned}$$

Preservation statement

Validity

$$\text{valid_tuple } (S, P) := \underline{(\text{dom } S) \cap (\text{vars_of } P) = \phi}$$

Matching solution

$$\text{match_sol } SI \ (S, P) := \frac{(s =_? t) \in P \rightarrow s =_{\sigma_{min}} t \langle SI \rangle \wedge \underline{\exists S', S \gg S' =_{\sigma_{min}} SI}}{}$$

Preservation theorem

$$\begin{aligned} \text{match_pres} : & \text{valid_tuple } T \rightarrow \\ & (\text{dom } SI) \cap (\text{lhvars_of } P) \rightarrow \\ & \text{smatch } T \ T' \rightarrow \\ & \text{match_sol } SI \ T' \rightarrow \\ & \text{match_sol } SI \ T \end{aligned}$$

Proof proceeds by case analysis on *smatch*

Inst case

$H_1 : (\text{dom } S) \cap \text{vars_of } P = \{\}$

$H_2 : (\text{dom } SI) \cap \text{lhvars_of } P = \{\}$

$H_3 : X \notin \text{lhvars_of } P$

$H_4 : \text{match_sol } SI (S \gg \{(X, s)\}, (P \setminus (s =_? X)) \langle \{(X, s)\} \rangle)$

$\text{match_sol } SI P$

Inst case (Unfolded *match_sol*)

$$H_1 : (\text{dom } S) \cap \text{vars_of } P = \{\}$$

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$$H_4 : \forall s' \ t', (s' =_? t') \in (P \setminus (s =_? X)) \langle \{(X, s)\} \rangle \rightarrow s =_{\sigma_{\min}} t \langle SI \rangle$$

$$H_5 : \exists S', SI =_{\sigma_{\min}} (S \gg \{(X, s)\}) \gg S'$$

$$\forall s_0, t_0, (s_0 =_? t_0) \in P \rightarrow s_0 =_{\sigma_{\min}} t_0 \langle SI \rangle \quad \wedge$$

$$\exists S', SI =_{\sigma_{\min}} S \gg S'$$

Inst case (Unfolded *match_sol*)

$$H_1 : (\text{dom } S) \cap \text{vars_of } P = \{\}$$

$$H_2 : (\text{dom } Sl) \cap \text{lhvars_of } P = \{\}$$

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Second conjunct is solvable associativity property of substitution composition.

Inst case continued..

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$$s_0, t_0 : \mathbf{exp}$$

$$H_6 : (s_0 =_? t_0) \in P$$

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Case analysis : $\{s_0 =_? t_0 = s =_? X\} + \{s_0 =_? t_0 \neq s =_? X\}$

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$s =_{\sigma_{min}} SI @ X$ is not immediate

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3. $(S \gg \{(X, s)\}) @ X = s$, from H_1 and H_6

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4. $((S \gg \{(X, s)\}) \gg S') @ X = ((S \gg \{(X, s)\}) @ X) \langle S' \rangle = s \langle S' \rangle$, from 3 and lookup property

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6. $s =_{\sigma_{min}} s \langle S' \rangle$, from 5

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Property mentioned in 5:

$\forall X\ S\ S'\ Sl, X \notin dom\ S \rightarrow X \notin Sl \rightarrow S \gg S' =_{\sigma_{min}} Sl \rightarrow S' @ X =_{\sigma_{min}} X$

Case 2.1: $s_0 =_? t_0 \neq s =_? X$, when $X \in \text{vars_of } t_0$

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1. We can show $s_0 =_? t_0 \langle \{(X, s)\} \rangle \in (P \setminus (s, X)) \langle \{(X, s)\} \rangle$.

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2. 1 in H_4 gives us $s_0 =_{\sigma_{\min}} t_0 \langle \{(X, s)\} \rangle \langle SI \rangle$

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3. $s_0 =_{\sigma_{\min}} t_0 \langle \{(X, s)\} \rangle \gg SI$, due to substitution composition property.
4. $Y \in (\text{vars_of } s) \rightarrow Y \notin \text{dom } SI$, from H_2 and H_6

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5. $SI @ X =_{\sigma_{\min}} s$, same as in case 1.

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3. $s_0 =_{\sigma_{\min}} t_0 \langle \{(X, s)\} \gg SI \rangle$, due to substitution composition property.
4. $Y \in (\text{vars_of } s) \rightarrow Y \notin \text{dom } SI$, from H_2 and H_6
5. $SI @ X =_{\sigma_{\min}} s$, same as in case 1.
6. $SI =_{\sigma_{\min}} \{(X, s)\} \gg SI$, from 4 and 5.

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1. We can show $\underline{s_0 =_? t_0 \langle \{(X, s)\} \rangle} \in (P \setminus (s =_? X)) \langle \{(X, s)\} \rangle$.
2. 1 in H_4 gives us $s_0 =_{\sigma_{\min}} t_0 \langle \{(X, s)\} \rangle \langle SI \rangle$
3. $s_0 =_{\sigma_{\min}} t_0 \langle \{(X, s)\} \rangle \gg SI$, due to substitution composition property.
4. $Y \in (\text{vars_of } s) \rightarrow Y \notin \text{dom } SI$, from H_2 and H_6
5. $SI @ X =_{\sigma_{\min}} s$, same as in case 1.
6. $SI =_{\sigma_{\min}} \{(X, s)\} \gg SI$, from 4 and 5.
7. $t_0 \langle SI \rangle =_{\sigma_{\min}} t_0 \langle \{(X, s)\} \rangle \gg SI$

Case 2.2: $s_0 =_? t_0 \neq s =_? X$, when $X \notin \text{vars_of } t_0$

$$H_1 : (\text{dom } S) \cap \text{vars_of } P = \{\}$$

$$H_2 : (\text{dom } SI) \cap \text{lhvars_of } P = \{\}$$

$$H_3 : X \notin \text{lhvars_of } P$$

$$H_4 : \forall s' t', (s' =_? t') \in (P \setminus (s =_? X)) \langle \{(X, s)\} \rangle \rightarrow s' =_{\sigma_{\min}} t' \langle SI \rangle$$

$$H_5 : SI =_{\sigma_{\min}} (S \gg \{(X, s)\}) \gg S'$$

$$s_0, t_0 : \text{exp}$$

$$H_6 : (s_0 =_? t_0) \in P$$

$$s_0 =_{\sigma_{\min}} t_0 \langle SI \rangle$$

1. Apply H_4 .
2. $(s_0 =_? t_0) \in (P \setminus (s, X)) \langle \{(X, s)\} \rangle$ is provable by knowing H_6 , $X \notin \text{vars_of } t_0$, and $s_0 \neq s$ and $t_0 \neq X$.